Today's agenda

- Intro to arrays
- Return sum of arr[] elements.
- Array with functions
- Swap 2 indexes.

AlgoPrep

// Intro to array

10 numbers

↳ int a1 = scn.nextInt();
int a2 = "
⋮
int a10 = "

1000 numbers
⇓
Array
⇓
Collection of data types (variables)

Syntax
→ type[ ] name = new type[size];

Q) Create an array of size 10 which will store integers.

↳ int[] arr = new int[10];

index of arr

arr

0	1	2	3	4	5	6	7	8	9
0	0	0	0	0	0	0	0	0	0

arr[2] arr[7]

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Indexing and Properties

↳ Scanner sch = new Scanner(System.in);

↳ int n = sch.nextInt(); → {5}

int[] arr = new int[n];

arr

0	1	2	3	4
0	0	0	0	0

→ for (int i=0; i<=n-1 ; i++) {

i < n
or

arr[i] = sch.nextInt();

}

arr

0	1	2	3	4
~~1~~ ~~0~~	~~2~~ ~~0~~	~~3~~ ~~0~~	~~4~~ ~~0~~	~~5~~ ~~0~~

i=0 i=1

→ if array is of length N.

1st index = 0

last index = N-1

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Q) Create an array of length 5 with values 10 20 30 40 50.

```
public static void main(String[] args) {
    Scanner scn = new Scanner(System.in);
    int n = scn.nextInt(); // n -> 5

    int[] arr = new int[n]; //arr -> {0, 0, 0, 0, 0}

    for(int i=0;i<n;i++){ // arr -> {10, 20, 30, 40, 50}
        arr[i] = scn.nextInt();
    }

    //System.out.println(arr);
     for(int i=0;i<n;i++){
         System.out.print(arr[i]+" ");
     }

    System.out.println(arr.length); // length of array

    // Initialization way 2
    int[] num = {5, 15, 25, 35, 45};
    System.out.println(num.length);

}
```

Q) Sum of array

↳ Read an array of N length and Print the sum of all elements.

Ex: arr[4]: 10 -1 3 -7 → 5

(0 1 2 3)

// Pseudo code

```
void main () {
    Scanner scn = new Scanner(System.in);
    int n = scn.nextInt();
    int[] arr = new int[n];
    for (int i=0; i<n; i++) {
        arr[i] = scn.nextInt();
    }
    int sum=0;
    for (int i=0; i<n; i++) {
        sum = sum + arr[i];
    }
    System.out.println(sum);
}
```

T.C: O(N)

S.C: O(1)

↓

length of array

(you ignore input & output array)

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```
int sum = 0;
for (int i=0; i<n; i++) {
    sum = sum + arr[i];
}
System.out.println(sum);
```

n=4

	0	1	2	3
arr[4]:	10	-1	3	-7

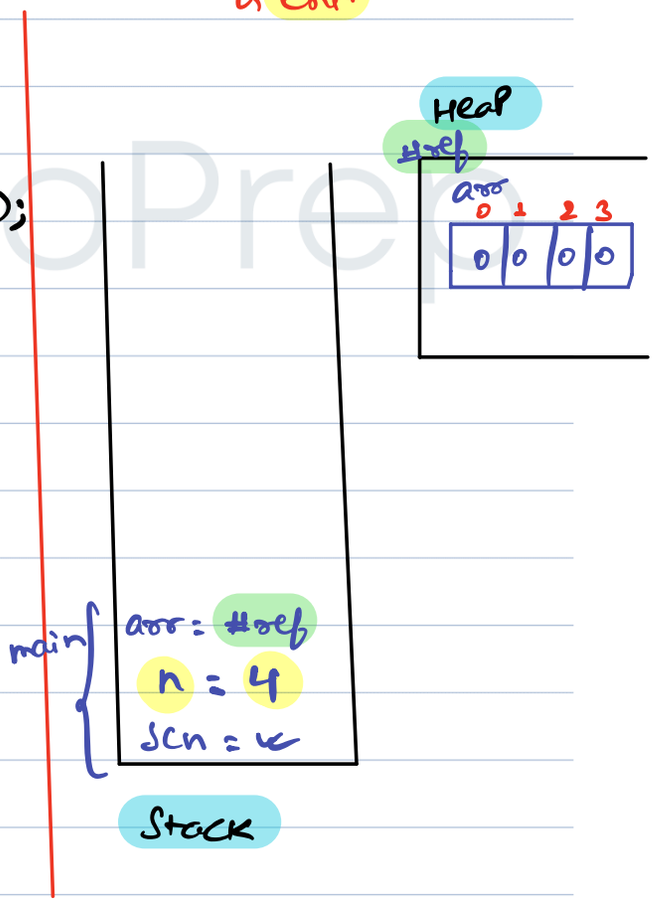
Sum = 0

i	i<n	sum
0	t	10
1	t	9
2	t	12
3	t	5
4	f ↳ exit	

Tracing

```
Scanner scn = new Scanner(System.in);
int n = scn.nextInt();
↳ int[] arr = new int[n];

System.out.println(arr[2]);
↳ 0
```



Break till 9:32 PM

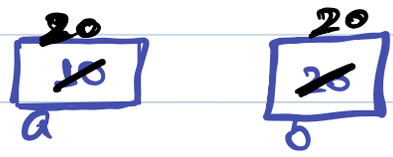
Q) Swap the values of 2 variables.

$a = 10$ $b = 20$ $\rightsquigarrow$ $a = 20$ $b = 10$

// incorrect way

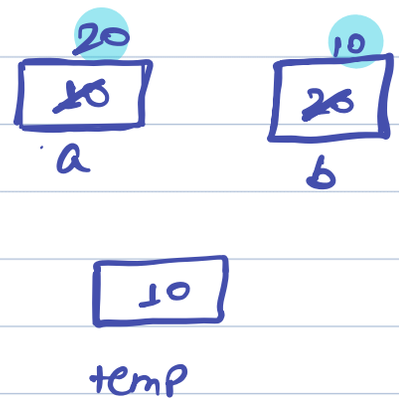
```
void main () {
    int a = 10;
    int b = 20;

    a = b;
    b = a;
}
```



// Correct way

```
void main () {
    int a = 10;
    int b = 20;
    int temp = a;
    a = b;
    b = temp;
}
```



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// function game

Quiz

```
main () {
    int a = 10;
    int b = 20;
    Swap(a, b);
    System.out.println(a);  → 10
    System.out.println(b);  → 20
}
```

```
Public static void Swap(int a, int b){
    int temp = a;
    a = b;
    b = temp;
}
```

Swap:
~~temp = 10~~
~~b = 20 10~~
~~a = 10 20~~

main:
b = 20
a = 10

↳ variables of 2 functions are not connected.

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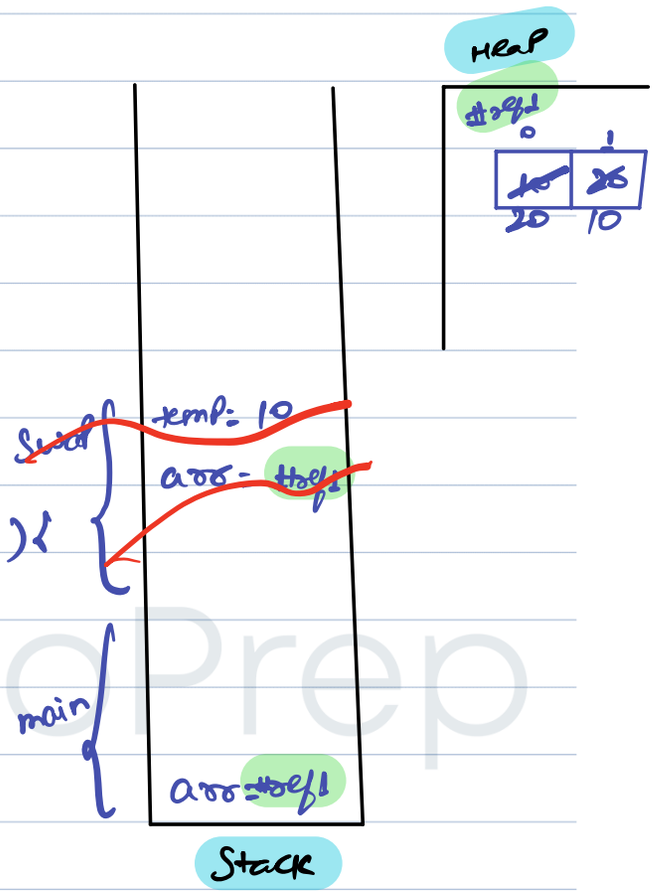
Quiz

```
main() {
    int[] arr = {10, 20}
    Swap(arr);
    System.out.println(arr[0]); → 20
    System.out.println(arr[1]); → 10
}

Public static void Swap( int[] arr ){
    int temp = arr[0];
    arr[0] = arr[1];
    arr[1] = temp;
}
```

#ref1



↳ arrays across functions are always connected.

Q) Swap indexes

↳ Given array of length N and two indexes idx1 and idx2, swap the element of those two indexes.

Ex: arr[5] : { 5 ~~15~~ 35 25 ~~25~~ 15 45 }

(indexes: 0, 1, 2, 3, 4)

idx1 = 1 idx2 = 3

```
int temp = arr[idx1];
arr[idx1] = arr[idx2];
arr[idx2] = temp;
```

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